Question 1: Bank Account Balance Calculator

AIM

To write a program to read the opening balance and deposit amount, validate both values, and display the resulting balance. Reject negative values.

ALGORITHM

Step 1: Start.

Step 2: Declare three variables:

- `OPEN_BALANCE` – to store the opening balance.
- `DEPOSIT_BALANCE` – to store the deposit amount.
- `UPDATED_BALANCE` – to store the final balance.

Step 3: Initialize all three variables to zero.

Step 4: Display the message "Enter open balance:".

Step 5: Read the opening balance from the user.

Step 6: Display the message "Enter deposit balance:".

Step 7: Read the deposit balance from the user.

Step 8: Check whether both the opening balance and deposit balance are negative.

- If both values are negative, display "Invalid Input".
- Otherwise, calculate the updated balance by adding the opening balance and deposit balance.

Step 9: Display the updated balance.

Step 10: Stop.

PROGRAM

```
IDENTIFICATION DIVISION.

PROGRAM-ID. HELLO-WORLD.

DATA DIVISION.

   WORKING-STORAGE SECTION.

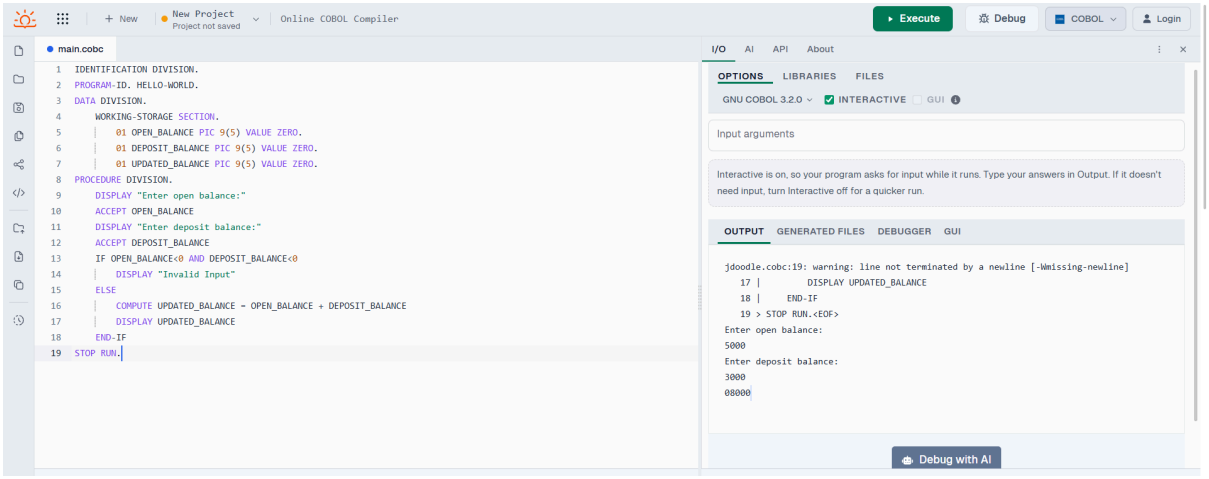
      01 OPEN_BALANCE PIC 9(5) VALUE ZERO.
```

```
        01 DEPOSIT_BALANCE PIC 9(5) VALUE ZERO.
        01 UPDATED_BALANCE PIC 9(5) VALUE ZERO.
PROCEDURE DIVISION.
    DISPLAY "Enter open balance:"
    ACCEPT OPEN_BALANCE
    DISPLAY "Enter deposit balance:"
    ACCEPT DEPOSIT_BALANCE
    IF OPEN_BALANCE<0 AND DEPOSIT_BALANCE<0
        DISPLAY "Invalid Input"
    ELSE
        COMPUTE UPDATED_BALANCE = OPEN_BALANCE + DEPOSIT_BALANCE
        DISPLAY UPDATED_BALANCE
    END-IF
STOP RUN.
```

OUTPUT

```
Enter open balance:
5000
Enter deposit balance:
3000
08000
```

SCREENSHOT


New
New Project
Project not saved
Online COBOL Compiler
Execute
Debug
COBOL
Login
main.cobc
1 IDENTIFICATION DIVISION.
2 PROGRAM-ID. HELLO-WORLD.
3 DATA DIVISION.
4 WORKING-STORAGE SECTION.
5 01 OPEN_BALANCE PIC 9(5) VALUE ZERO.
6 01 DEPOSIT_BALANCE PIC 9(5) VALUE ZERO.
7 01 UPDATED_BALANCE PIC 9(5) VALUE ZERO.
8 PROCEDURE DIVISION.
9 DISPLAY "Enter open balance:"
10 ACCEPT OPEN_BALANCE
11 DISPLAY "Enter deposit balance:"
12 ACCEPT DEPOSIT_BALANCE
13 IF OPEN_BALANCE<0 AND DEPOSIT_BALANCE<0
14 DISPLAY "Invalid Input"
15 ELSE
16 COMPUTE UPDATED_BALANCE = OPEN_BALANCE + DEPOSIT_BALANCE
17 DISPLAY UPDATED_BALANCE
18 END-IF
19 STOP RUN.
I/O
AI
API
About
OPTIONS
LIBRARIES
FILES
GNU COBOL 3.2.0
INTERACTIVE
GUI
Input arguments
Interactive is on, so your program asks for input while it runs. Type your answers in Output. If it doesn't need input, turn Interactive off for a quicker run.
OUTPUT
GENERATED FILES
DEBUGGER
GUI
jdoodle.cobc:19: warning: line not terminated by a newline [-Wmissing-newline]
17 | DISPLAY UPDATED_BALANCE
18 | END-IF
19 > STOP RUN.<EOF>
Enter open balance:
5000
Enter deposit balance:
3000
08000
Debug with AI